

Hubs, Bridges, and LAN Switches

Repeaters and hubs extend a collision domain: every frame is flooded to every port. Bridges and switches break that domain. A transparent switch maintains a MAC address table, learns source addresses from incoming frames, and forwards unicast traffic only to the destination port. Unknown unicast, broadcast, and multicast frames are still flooded inside the broadcast domain.

Store-and-forward switching checks the frame check sequence before forwarding, which discards damaged frames. Cut-through switching starts forwarding earlier and has lower latency, but it can propagate errors. Most campus switches default to store-and-forward.

Layer-2 loops are dangerous. Spanning Tree Protocol blocks redundant links until a failure occurs. Rapid spanning tree converges faster than the classic algorithm. Administrators should enable portfast or edge-port features only on access ports that connect end hosts, never on inter-switch uplinks.

Replacing hubs with switches is the single most effective upgrade for an aging LAN. Each workstation receives a dedicated collision domain, full-duplex Ethernet becomes possible, and capture of other users' traffic requires a configured mirror port rather than a cheap hub.